

Olist E-Commerce Dataset

Customer Cohort Analysis and Retention Report

Prepared for Data Analysis Internship Submission

Prepared by Data Analyst | Industry-style Analytical Report

Executive Summary

Total Customers Analysed 69,127
Repeat Customers 1,590 (2.30%)
One-Time Customers 67,537 (97.70%)
Total Orders Analysed 82,365
Average Customer Lifetime Value R\$ 118.91
Average Delivery Duration 11.97 days
Delayed Orders 5,240 (6.36%)
Cohort Period Analysed Sep 2016 to Aug 2018

Customer retention is the primary business challenge. While acquisition is strong, 97.7% of customers purchase only once.

1. Scope, Objective and Analytical Approach

Objective: Identify customer retention patterns, cohort behaviour, and churn drivers from the Olist e-commerce transaction dataset to support marketing, retention, and operational decision-making.

Analytical grain: Customer analysis is performed at customer_unique_id level; cohort analysis is at monthly cohort level; product category analysis uses product_category_name_english. All analyses are restricted to delivered and active orders for reliable revenue and retention interpretation.

Why cohort analysis: Standard aggregate retention metrics mask the timing of customer drop-off. Monthly cohort analysis reveals exactly when customers stop returning and enables comparison across acquisition periods.

2. Dataset Overview and Preparation

The dataset is a cleaned master e-commerce dataset combining customer, order, product, payment, seller, and delivery information. All columns were verified, date fields were converted to datetime format, and derived features including OrderMonth, CohortMonth, CohortIndex, delivery_days, delivery_delay, and delivery_status were engineered prior to analysis.

Table 1. Dataset Summary

Metric	Value	Notes
Total Rows	82,365	One row per order
Total Columns	39	Including engineered features
Unique Customers	69,127	customer_unique_id level
Cohort Period	Sep 2016 to Aug 2018	Sep/Oct 2018 are incomplete
Delivered Orders	79,947 (97.06%)	Primary analysis base
Cancelled/Unavailable	998 (1.21%)	Excluded from revenue metrics

3. Customer Cohort Retention Analysis

The cohort retention matrix groups customers by the month of their first purchase (cohort month) and tracks how many of those customers returned in each subsequent month (cohort index). The analysis covers 25 monthly cohorts from September 2016 to August 2018.

Table 2. Cohort Customer Count Summary (First-Month Acquisition Size)

Cohort Month	Customers Acquired	Avg Retention %	Total Revenue (R\$)
2016-09	4	100.00%	207.86
2016-10	228	25.33%	28,464.63
2017-01	539	9.33%	76,651.61

2017-02	1,247	6.43%	144,266.38
2017-03	1,913	5.71%	244,247.12
2017-04	1,686	6.83%	229,459.72
2017-05	2,560	6.47%	338,920.46
2017-06	2,271	6.85%	262,676.89
2017-07	2,808	7.33%	304,753.75
2017-08	3,006	7.90%	346,747.29
2017-09	2,934	9.30%	383,585.14
2017-10	3,223	10.17%	401,000.97
2017-11	5,282	10.16%	611,761.72
2017-12	3,923	11.25%	445,228.73
2018-01	5,062	12.67%	563,521.06
2018-02	4,649	14.48%	518,032.89
2018-03	4,967	16.84%	566,273.97
2018-04	4,781	20.21%	586,063.54
2018-05	4,807	25.19%	585,909.54
2018-06	4,253	33.53%	530,056.16
2018-07	4,430	33.55%	539,076.68
2018-08	4,543	33.36%	512,827.36

Note: Sep 2018 (8 customers) and Oct 2018 (2 customers) are incomplete data months and are excluded from trend interpretation.

3.1 Retention Analysis Findings

The cohort retention analysis reveals a consistent and significant pattern across all acquisition cohorts: customer activity concentrates heavily in the first month, with very limited repeat engagement in subsequent periods.

- Early cohorts (Sep 2016 to Jun 2017) show average retention rates between 5.71% and 25.33%, indicating that the vast majority of acquired customers did not return.
- Mid cohorts (Jul 2017 to Dec 2017) show marginal improvement, with average retention rates between 7.33% and 11.25%, though still extremely low by industry standards.
- Later cohorts (Jan 2018 to Aug 2018) show progressively improving average retention rates from 12.67% to 33.55%. However, this improvement is partly a dataset artifact: newer cohorts have had less time to churn, so their retention figures appear higher because fewer follow-up months are observed.
- The November 2017 cohort is the largest single acquisition cohort at 5,282 customers and generated the highest cohort revenue of R\$ 611,761.72.

3.2 Retention Rate Matrix Interpretation

The retention heatmap produced from the analysis visualises customer persistence across cohort months and cohort index (months since first purchase). Key observations from the matrix include:

- Month 1 retention is 100% by construction for all cohorts, as it represents the acquisition month itself.
- Month 2 retention (first return) drops to near-zero across most cohorts, with only sparse repeat purchase activity observed.
- Month 3 and beyond show continued decline, with retention rates well below 5% for most early cohorts.
- The retention heatmap is predominantly empty (NaN), confirming that the vast majority of customers make only a single purchase with no follow-up activity.

4. Customer Churn Analysis

Churn is defined as the complement of retention: customers who do not return in a given cohort month. The churn matrix mirrors the retention matrix, with values approaching 100% in all periods beyond Month 1 for most cohorts.

Table 3. Churn Rate Summary by Cohort Period

Cohort Period	Approximate Churn After Month 1	Business Interpretation
Sep 2016 - Dec 2016	75% to 100%	Earliest cohorts; very high churn
Jan 2017 - Jun 2017	~93% to 95%	Growth phase; retention remains poor
Jul 2017 - Dec 2017	~88% to 93%	Peak revenue period; churn still high
Jan 2018 - Aug 2018	~67% to 88%	More observed data; apparent improvement

The churn heatmap confirms that customer churn is a structural issue across the entire dataset period and is not isolated to any specific acquisition cohort or time of year. The business loses approximately 90 to 97 out of every 100 acquired customers after their first purchase.

5. Revenue Cohort Analysis

Revenue cohort analysis tracks the total product revenue generated by each cohort across purchase months. This reveals which cohorts are most commercially valuable and how cohort revenue evolves over time.

Table 4. Top Revenue-Generating Cohorts

Cohort Month	Customers	Total Revenue (R\$)	Revenue Rank
2017-11	5,282	611,761.72	#1
2018-04	4,781	586,063.54	#2

2018-05	4,807	585,909.54	#3
2018-01	5,062	563,521.06	#4
2018-03	4,967	566,273.97	#5
2018-06	4,253	530,056.16	#6
2018-07	4,430	539,076.68	#7
2018-08	4,543	512,827.36	#8

The November 2017 cohort generated the highest total revenue of R\$ 611,761.72, driven by its large acquisition size of 5,282 customers. Revenue cohorts in 2018 are consistently strong due to high monthly acquisition volumes, but this revenue is almost entirely first-purchase driven. Without improving repeat purchase rates, cohort revenue will remain acquisition-dependent.

6. Monthly Revenue Trend

Monthly revenue was computed by summing all product prices within each calendar month, regardless of cohort membership. This provides an overall view of business revenue trajectory.

Table 5. Monthly Revenue Trend (Selected Months)

Month	Revenue (R\$)	Trend Note
2017-01	75,760.07	Early growth phase
2017-06	257,026.69	Steady mid-year growth
2017-11	608,903.06	Peak month (Nov 2017)
2018-01	567,610.38	Post-peak sustained level
2018-04	590,345.27	2018 revenue peak
2018-05	589,159.17	Slight decline begins
2018-06	541,620.93	Declining trend
2018-08	520,460.17	Dataset ends; trend unclear

Revenue grew steadily from 2016 through November 2017 (peak at R\$ 608,903). A plateau and mild decline are observed from mid-2018 onward. Since almost all revenue comes from first-time purchases, this decline reflects a reduction in new customer acquisition rates rather than a collapse in existing customer spending.

7. Repeat Customer and Customer Lifetime Value Analysis

7.1 Repeat vs One-Time Customer Distribution

Table 6. Customer Purchase Frequency

Customer Type	Count	Share (%)
One-Time Customers	67,537	97.70%
Repeat Customers	1,590	2.30%
Total Unique Customers	69,127	100%

97.70% of all customers in the dataset made only a single purchase. Only 2.30% returned for at least one additional transaction. This is the most significant finding of the analysis and represents a fundamental retention challenge for the business.

7.2 Customer Lifetime Value (CLTV)

Table 7. Customer Lifetime Value Distribution

Metric	One-Time Customers	Repeat Customers
Average Revenue (R\$)	116.44	223.69
Overall Mean CLTV (R\$)	118.91 (all customers)	—
Median CLTV (R\$)	66.99 (all customers)	—
Maximum CLTV (R\$)	13,440.00 (top spender)	—

Repeat customers generate on average 92% more revenue than one-time customers (R\$ 223.69 vs R\$ 116.44). The heavily right-skewed CLTV distribution (mean R\$ 118.91 vs median R\$ 66.99) suggests that a small number of high-spending customers drive a disproportionate share of revenue. Improving repeat purchase rates would directly increase average CLTV and total revenue.

8. Root Cause Analysis: Repeat Customer Decline

Since the dataset does not contain direct customer feedback or exit survey data, the reasons for low repeat purchase rates were identified indirectly through behavioural and operational data analysis across six dimensions.

8.1 Delivery Performance Analysis

Table 8. Delivery Performance: Repeat vs One-Time Customers

Metric	One-Time Customers	Repeat Customers
Avg Delivery Days	12.02 days	11.67 days
Avg Delivery Delay	-11.88 days (early)	-12.52 days (early)
Total Delayed Orders	3,400 (delayed)	111 (delayed)
Delayed Order Rate	6.36% of total orders	6.36% overall

The analysis does not show a strong differentiation in delivery performance between repeat and one-time customers, indicating that delivery delays alone do not fully explain the high churn rate. However, 5,240

delayed deliveries (6.36% of all orders) represent a persistent operational risk that may contribute to marginal customer dissatisfaction.

8.2 Product Category Repeat Rate Analysis

Product category analysis identifies whether the composition of the product catalogue itself drives low repeat rates. Categories with naturally infrequent purchase cycles will inherently produce lower repeat customer rates regardless of customer satisfaction.

Table 9. Top 10 Product Categories by Repeat Purchase Rate

Product Category	One-Time	Repeat	Repeat Rate %
fashio_female_clothing	22	6	21.43%
arts_and_craftmanship	13	3	18.75%
home_appliances	402	66	14.10%
drinks	209	32	13.28%
fashion_sport	16	2	11.11%
furniture_living_room	275	33	10.71%
fashion_bags_accessories	1,108	118	9.62%

Table 10. Bottom 10 Product Categories by Repeat Purchase Rate (Zero or Near-Zero Repeat)

Product Category	One-Time	Repeat	Repeat Rate %
costruction_tools_garden	138	1	0.72%
christmas_supplies	85	1	1.16%
computers	127	2	1.55%
flowers	19	0	0.00%
cds_dvds_musicals	9	0	0.00%
security_and_services	2	0	0.00%
fashion_childrens_clothes	6	0	0.00%

Categories such as construction tools, computers, flowers, and security services are structurally one-time purchase items. The dominance of these low-repurchase-frequency categories in the Olist catalogue is a significant contributor to the overall low repeat purchase rate. In contrast, fashion, home appliances, and drinks show relatively higher repeat rates (10% to 21%), suggesting these categories could be prioritised in retention campaigns.

8.3 Order Cancellation and Payment Analysis

Table 11. Payment Type: One-Time vs Repeat Customers

Payment Type	One-Time Customers	Repeat Customers
credit_card	39,227	1,470
boleto	10,358	338
voucher	2,938	126
debit_card	774	30

Credit card is the dominant payment method for both customer types. Voucher usage is relatively low but may represent customers who responded to promotional campaigns. Cancellation rate analysis shows that repeat customers had an average of 4.47% cancelled orders vs 1.19% for one-time customers, suggesting that customers who continue shopping are also more likely to encounter inventory or availability issues.

8.4 Average Order Value Comparison

Table 12. Average Order Value: One-Time vs Repeat Customers

Customer Type	Avg Revenue Per Customer (R\$)	Insight
One-Time Customers	116.44	Lower total spend; single visit
Repeat Customers	223.69	Nearly 2x higher lifetime value

Repeat customers generate 92% more revenue per customer than one-time customers. This clearly quantifies the business value of improving retention: converting even a fraction of one-time buyers into repeat customers would significantly increase total revenue without requiring additional customer acquisition spend.

8.5 Time-to-Second Purchase Analysis

Table 13. Days to Second Purchase Statistics

Statistic	Value	Interpretation
Count (second purchasers)	5,621 records	Multiple orders from repeat buyers
Mean Days to 2nd Purchase	9.57 days	Fast second purchase if it occurs
Median Days to 2nd Purchase	0.00 days	Many same-day or bundled orders
Maximum Days to 2nd Purchase	517 days	Some very delayed second purchases

The median of 0 days indicates that most second purchases occur on the same day or very close to the first, suggesting that these are bundled or split orders rather than genuine returning customers. The mean of 9.57 days confirms that genuine repeat purchase cycles in this dataset are extremely short, and customers who do not return immediately are very unlikely to return later.

8.6 Shipping Cost Analysis

Table 14. Shipping Cost Ratio Statistics

Metric	Value	Interpretation
Average Shipping Ratio	26.41%	Avg freight is 26% of product price
Median Shipping Ratio	23.30%	Typical order pays ~23% in freight
75th Percentile	39.43%	25% of orders pay over 39% in freight
Maximum Ratio	2,623%	Extreme outliers: freight > product price

An average shipping cost of 26.41% of product value is high by e-commerce standards, where free or low-cost shipping is increasingly the expectation. For low-value product categories, freight charges may approach or exceed the product price, significantly reducing the perceived value of re-purchasing. This is a likely contributing factor to one-time purchase behaviour.

9. Geographic Retention Analysis

State-level retention analysis identifies whether certain regions show systematically lower repeat purchase rates, which could indicate logistics, delivery, or market-specific issues.

Table 15. Top States by Revenue and Repeat Purchase Rate

State	One-Time	Repeat	Total	Repeat Rate %	Revenue (R\$)
SP	32,818	1,834	34,652	5.29%	3,125,614
RJ	10,206	604	10,810	5.59%	1,117,654
MG	8,995	473	9,468	5.00%	947,023
RS	4,229	242	4,471	5.41%	444,413
PR	3,971	206	4,177	4.93%	417,990
BA	2,747	130	2,877	4.52%	320,978
SC	2,901	133	3,034	4.38%	317,552
GO	1,627	74	1,701	4.35%	189,863
DF	1,642	67	1,709	3.92%	174,282

Repeat purchase rates are consistently low across all major states, ranging from 3.92% (DF) to 5.59% (RJ). This uniformity suggests that low retention is a platform-wide structural issue rather than a region-specific operational problem. No single state shows meaningfully higher repeat purchase engagement.

10. Key Findings Summary

Table 16. Prioritised Key Findings

#	Finding Area	Finding	Business Impact
1	Customer Retention	97.70% of customers make only one purchase; only 2.30% return	Very High
2	Customer Lifetime Value	Repeat customers generate 92% more revenue (R\$ 223.69 vs R\$ 116.44)	Very High
3	Revenue Peak	Nov 2017 was the highest revenue month at R\$ 608,903; revenue plateaued in 2018	High
4	Product Category Mix	Many high-volume categories have 0-2% repeat rates (computers, construction, flowers)	High
5	Shipping Costs	Average freight = 26.41% of product value; 25% of orders exceed 39%	Moderate
6	Delivery Performance	6.36% of orders delayed; no significant retention difference between customer types	Moderate
7	Geographic Uniformity	Retention is uniformly low across all major states (3.9% to 5.6%)	Moderate
8	Purchase Timing	Customers who do not repurchase on first visit are unlikely to return (median 0 days to 2nd purchase)	Moderate

11. Business Recommendations

Table 17. Prioritised Recommendations

Area	Recommendation	Priority
Customer Retention	Launch structured post-purchase re-engagement campaigns targeting customers within 7 to 30 days of first purchase, before the critical drop-off window.	Very High
Loyalty Programme	Introduce a loyalty points or rewards programme to incentivise second and third purchases, particularly in high-potential categories such as fashion, home appliances, and drinks.	Very High
Product Mix Strategy	Expand catalogue share in high-repeat-rate categories (fashion_female_clothing, arts_and_craftmanship, home_appliances, drinks) and reduce promotional spend on structural one-time categories.	High
Shipping Cost Optimisation	Introduce free shipping thresholds or subsidised shipping for returning customers to reduce the 26.41% average freight burden and improve perceived purchase value.	High
Logistics Improvement	Address the 6.36% delayed delivery rate through carrier SLA reviews and proactive customer communication to prevent satisfaction erosion.	Moderate
Customer Segmentation	Segment customers into high-value returners, potential returners, and dormant one-time buyers to enable differentiated marketing spend and channel allocation.	Moderate

Regional Expansion	SP dominates revenue (37% share in the reference analysis). Target underpenetrated high-population states with tailored acquisition campaigns and delivery SLA improvements.	Moderate
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12. Limitations and Next Steps

The following limitations apply to this analysis and should be considered when interpreting findings:

- The dataset covers September 2016 to August 2018. September and October 2018 contain only 8 and 2 customers respectively and are treated as incomplete data months excluded from trend conclusions.
- Profit and margin data are not available; revenue-driving categories and cohorts may not be the most profitable after seller commissions and operating costs.
- Customer demographic attributes (age, gender, income) are not present. Retention segmentation is therefore behaviour-based only.
- The near-zero median days to second purchase (0 days) suggests that many repeat orders are bundled or split deliveries rather than genuine returning customer visits.
- Causal inference is not possible from transactional data alone. Findings on root causes of churn represent correlation-based indications, not confirmed causal relationships.

Recommended next steps include cohort-based churn prediction modelling, market basket analysis to identify cross-sell opportunities, seller performance scorecarding, and A/B testing of retention campaign designs using the customer segments identified in this analysis.

Appendix A. Analysis Outputs Generated

Output File	Description
customer_retention_matrix.csv	Monthly cohort retention counts (customers x cohort index)
retention_rate_matrix.csv	Retention rate matrix normalised by cohort size
churn_rate_matrix.csv	Churn rate matrix (1 minus retention rate)
revenue_cohort_matrix.csv	Revenue by cohort month and cohort index
cohort_summary.csv	Summary of cohort size, average retention, and total revenue
customer_lifetime_value.csv	Revenue per unique customer across full transaction history
product_category_repeat_rate.csv	Repeat purchase rate by product category

All analyses were performed using Python (pandas, numpy, matplotlib, seaborn) in a Jupyter Notebook environment using the analysis_data.csv cleaned master dataset.